ENGINEERING PORTFOLIO · 2026

Enrique Valverde Sacristán

MSc Aerospace Engineering student, ISAE-SUPAERO (Aerospace Systems & Control)
Mechanical engineer

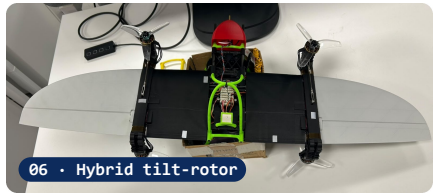
Model-based state estimation and UAV control, evaluated with Monte Carlo campaigns and real flight data. Looking to push estimation and flight-control performance on fast, real hardware.

LOOKING FOR

UAV flight control, GNC and state estimation roles
Full-time role or six-month final-year placement, from **March 2027**


08 · Interceptor UAV


05 · Real-flight test


06 · Hybrid tilt-rotor

03 INOAS: navigation with an intermittent GNSS receiver
UKF, pseudo-NIS sensor-fault alarm, NEES/NIS consistency, real GNSS error data · IEEE Aerospace 2027 (abstract accepted)

05 From simulation to flight: PID vs INDI
Real quadrotor flight logs, OptiTrack ground truth, error decomposition

06 INDI control of a hybrid tilt-rotor UAV
ENAC 6-DoF simulator, WLS allocation, tether sensitivity study

07 Sunspear: estimation for rocket airbrakes
IMU-driven Kalman filter, regime-scheduled sensors, MPC

08 Low-cost interceptor UAV
Camera/LiDAR mount, avionics, Betaflight; field test with the end user · team 1st place

09 Real-flight labs
Adaptive control with payload release; onboard sensing vs motion capture



CONTACT

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Toulouse, France · Spanish nationality

AT A GLANCE

MSc	**ISAE-SUPAERO**, Aerospace Systems & Control (2025–2027)
BSc	Mechanical Engineering, UC3M · Erasmus+ dual-degree programme at Arts et Métiers Paris
Industry	6-month internship, Safran Aircraft Engines
UAS	EASA Open A1/A3 and A2 remote-pilot certificates
Languages	English C1 (C1 Advanced, IELTS 7.5) · French B2 · Spanish native

ISAE Institut Supérieur de l'Aéronautique et de l'Espace SUPAERO

WHERE EACH SKILL SHOWS UP IN MY WORK

Six projects at a glance

State estimation and flight control, backed by simulation and real flight-data analysis.

SKILL	INOAS 03–04	Flight test 05	Tilt-rotor 06	Sunspear 07	Interceptor 08	Labs 09
Real sensor and flight data logs, ground truth, error analysis	●	●				●
State estimation and sensor fusion Kalman filters, NEES / NIS, sensor scheduling	●	○	○	●		○
GPS-denied and degraded sensors GNSS off, biased sensors, IMU-only phases	●			●		
Sensor timing and latency sync filters, sensor lag, sample rates			●			●
Flight control INDI, PID, MRAC, MPC	○	●	●	○		●
Models and simulation 6-DoF simulators, Monte Carlo, sensor models	●	○	●	●		○
Hardware and flight tests integration, field and indoor flights		●	○		●	●

● core of the project ○ secondary

INDUSTRY **Safran Aircraft Engines, 2024.** Six-month internship: finite-element models of a LEAP fan-case bond, kept in use by the design office after I left.

IEEE
Aerospace Conference 2027, 2nd author (abstract accepted)

250
paired Monte Carlo runs behind the INOAS results

10
real flights analysed against motion capture

1st
team place of four, low-cost drone challenge

WHAT I WOULD BRING

Model-based estimation, controller evaluation and flight-log analysis. Now helping port a Kalman filter to C/FreeRTOS; next, real-time flight software in Rust.

TOOLBOX

KALMAN / UKF · NEES / NIS · GNSS ERROR MODELLING · MPC · INDI · WLS ALLOCATION · PYTHON · MATLAB / SIMULINK · GIT · FLIGHT-LOG ANALYSIS · C / FREERTOS · BETAFLIGHT

SPACE GNC · STUDENT AEROSPACE CHALLENGE → IEEE AEROSPACE CONFERENCE 2027 (2ND AUTHOR, ABSTRACT ACCEPTED)

INOAS: navigating with a GNSS receiver that is mostly off

A LEO CubeSat that avoids debris on its own while its GNSS receiver sleeps **70 % of the time**.

PROBLEM

An always-on GNSS receiver drains a CubeSat. Switched off, the navigation error grows.

→

APPROACH

UKF + supervisor that wakes the receiver on a sensor-fault alarm + MPC whose keep-out grows with uncertainty.

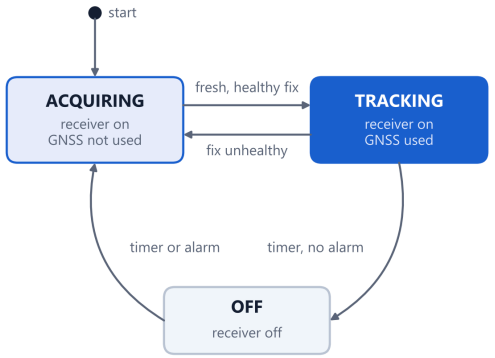
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RESULT

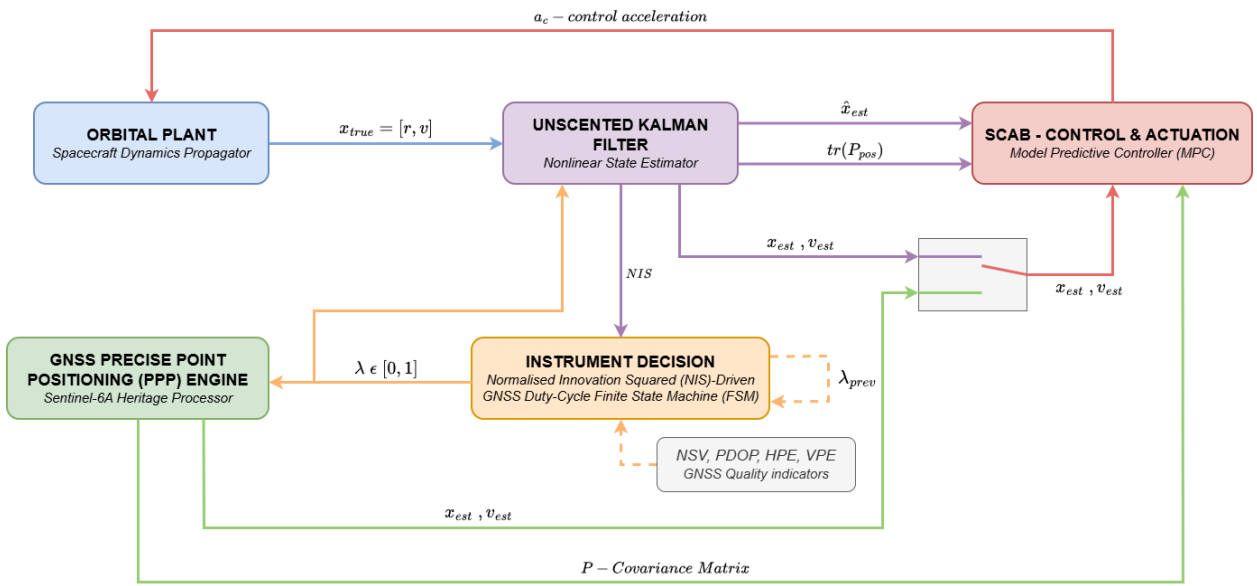
Median max error 8.0 m vs 21.1 m with a fixed timer; −66 % modelled receiver energy; no keep-out violation in 250 simulated runs.

MY PART

- UKF navigation, its process-noise budget and the three-state receiver supervisor.
- MPC guidance whose keep-out grows with the navigation covariance.
- Monte Carlo campaigns and all paper figures.



Receiver states. OFF → ACQUIRING → TRACKING; an alarm or a timer wakes it up.



Challenge architecture. Navigation quality decides when to power the receiver; the navigation covariance sets the MPC keep-out radius.

INOAS · EVALUATION OVER 250 PAIRED MONTE CARLO RUNS

Judging a filter by its consistency, not only its error

Bounded error, less receiver energy, and a covariance that is **no longer overconfident**.

8.0 vs 21.1 m
median max position error, reactive vs fixed timer (lower in 50/50 pairs)

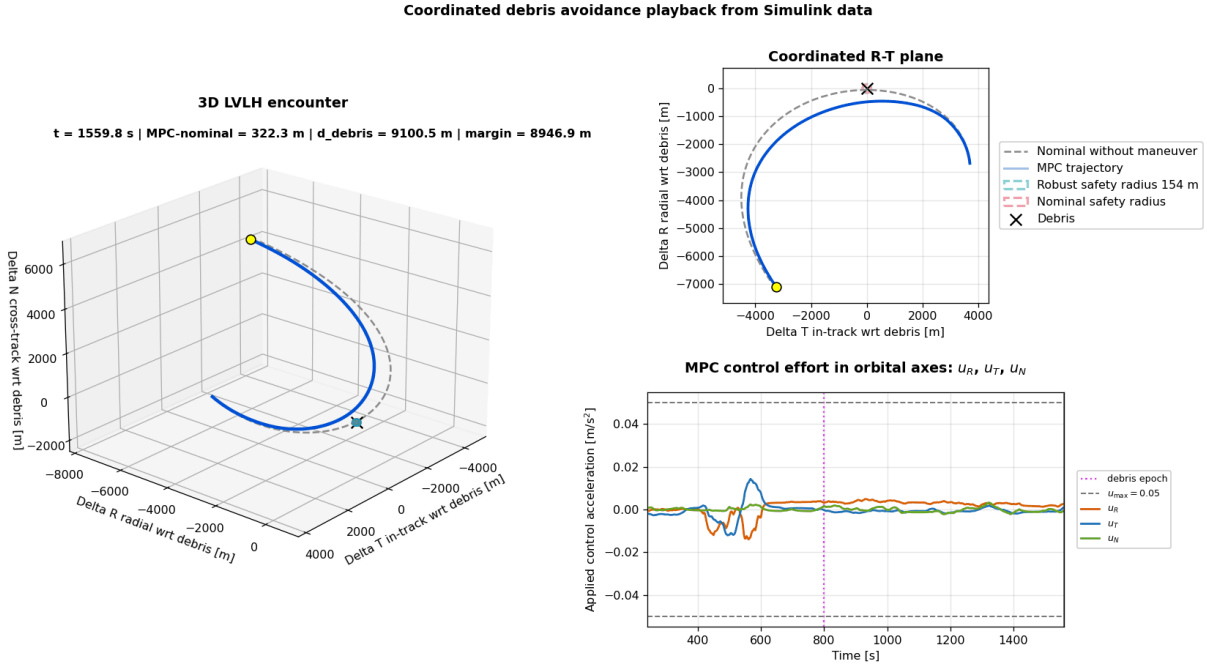
3–7 s
to flag an injected +5 km sensor bias

−32 %
Δv, covariance-adaptive vs constant 295 m keep-out

−66 %
modelled receiver energy vs always-on GNSS

12,650 → 0.23 NEES, expected 6
previous vs budget-based process noise (omitted forces): no longer overconfident; conservative, not calibrated (single run)

0.58 m RMS
real Sentinel-6A GNSS errors, but correlated for minutes



CHALLENGE PLAYBACK

Debris avoidance with MPC

The MPC trajectory against the nominal path without a manoeuvre, in 3D and in the radial/in-track plane, with the control effort in orbital axes well within its ±0.05 m/s² limit.

Earlier Challenge configuration, not the 250-run evaluation above.

UAV FLIGHT DATA · ISAE-SUPAERO × ENAC RESEARCH PROJECT · 31 MARCH 2026

From simulation to flight: PID vs INDI

In flight, most of the error was **one offset shared by both controllers**.

QUESTION

Does INDI's advantage in simulation survive real flight data?

→

TEST

Same 2 m circle and metrics in simulation and in the ENAC arena, OptiTrack as ground truth.

→

FINDING

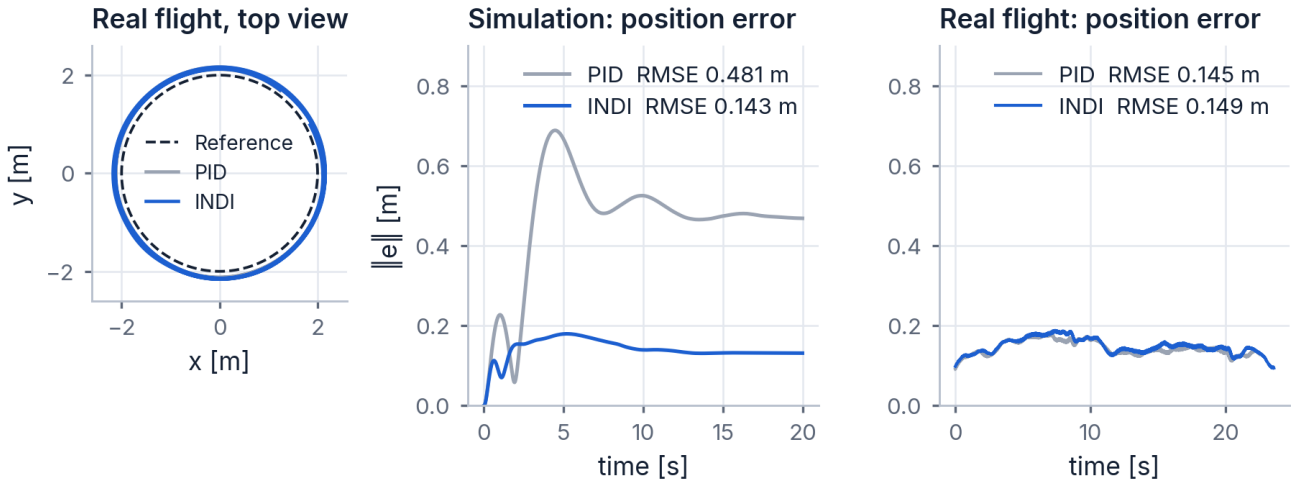
Sim: 70 % lower error with INDI. Flight: both ≈ 15 cm, mostly one shared 14 cm outward offset.

MY PART

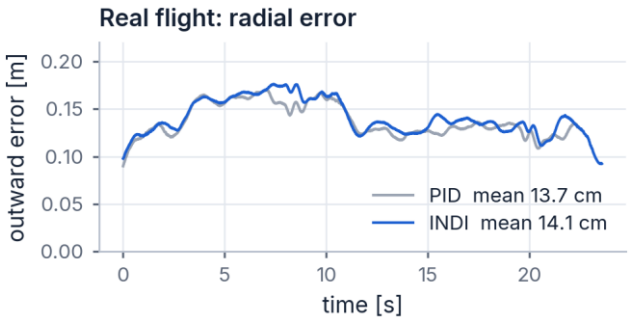
- Wrote the baseline flown in the test (a PD + P cascade).
- Wrote the log-processing and metrics pipeline.
- Error decomposition that exposed the shared offset.



ENAC flight arena with OptiTrack cameras.



Same reference, same metrics. Flown circle, then position error in simulation and in flight on the same scale.



TAKEAWAY

Similar radial offsets in both flights suggest a shared cause (calibration, frame alignment or thrust mapping).

One flight per controller cannot isolate it: an implementation check, not a ranking.

FLIGHT CONTROL & SIMULATION · ISAE-SUPAERO × ENAC RESEARCH PROJECT · SINCE FEB 2026

INDI control of a hybrid quad-tiltrotor, free and tethered

Accurate on the nominal orbit; a ±15 % sweep showed **where it breaks**.

PROBLEM
Over-actuated tilt-rotor (4 rotors, 2 tilts, 1 wing) pulled by a tether that goes taut and slack.

→

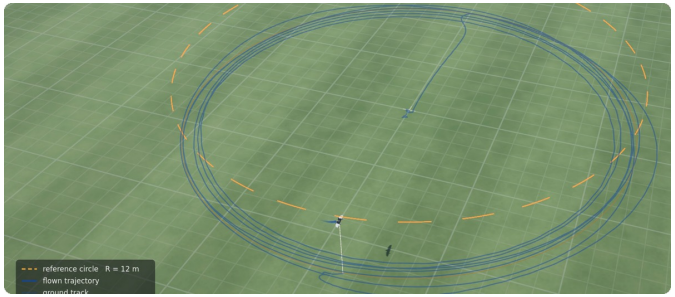
APPROACH
Cascaded INDI with WLS allocation on ENAC's 6-DoF simulator; one-at-a-time ±15 % sweep.

→

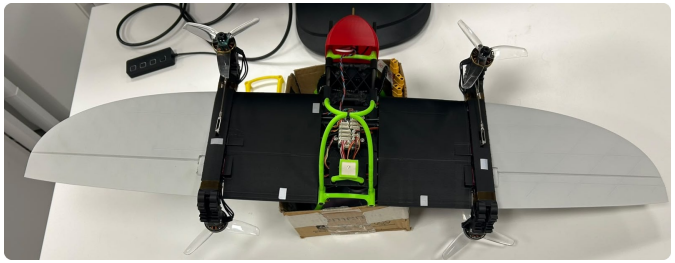
RESULT
8.7 cm RMSE on the nominal tethered orbit; 8 of 16 perturbations diverge.

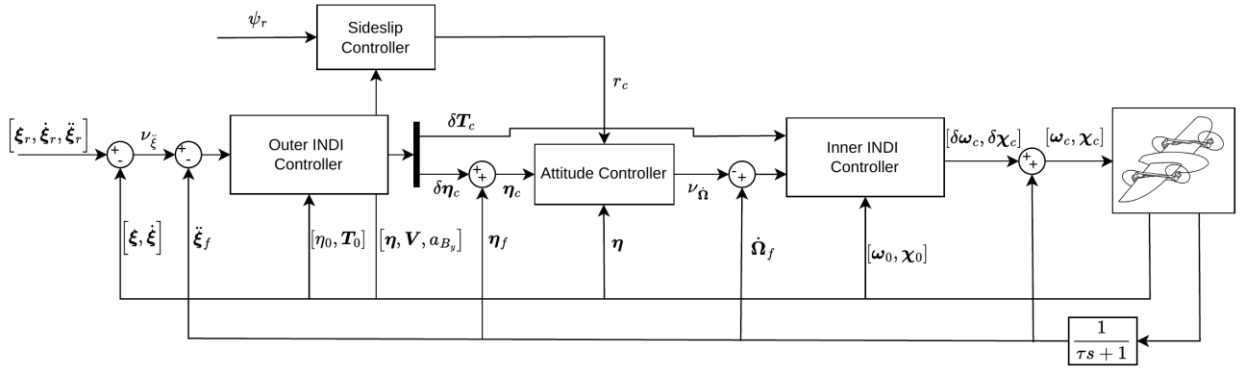
MY PART

- Found the INDI synchronisation filter effectively off ($\tau = 0$) in the simulator.
- Modelled the flexible tether and ran its sensitivity campaign (standalone report).
- Implemented the constrained WLS allocation.



Outdoor flight with this controller, 22 Oct 2026 (replay of the GNSS log).
Below: the hybrid quad-tiltrotor on the bench.





Cascaded INDI. 100 Hz guidance and 200 Hz stabilisation loops, each allocating with constrained WLS.

	−15 %	+15 %
Cable stiffness k_c	diverges	diverges
Cable damping d_c	RMSE 11 cm	RMSE 8 cm
Cable length L_c	diverges	diverges
Linear density μ	diverges	diverges
Cable diameter	RMSE 11 cm	RMSE 8 cm
Segments N	RMSE 26 cm	diverges
UAV mass m	diverges	RMSE 10 cm
Orbit speed v_{max}	RMSE 6 cm	RMSE 44 cm

WHY IT FAILS

The orbit sits at the taut/slack boundary.

With almost no pre-tension, a perturbed UAV loses altitude, the cable goes slack, then snaps taut (198 N peaks in the stiffness case).

Tools Python · NumPy/SciPy

github.com/enriquev212/hybrid-tiltrotor-indi-control

STATE ESTIMATION · SUPAERO SPACE SECTION · SINCE SEPTEMBER 2025

Sunspear: estimation for a supersonic rocket's airbrakes

Baro and GNSS become unreliable in flight, so the filter **picks its sensors by flight regime**.

PROBLEM
Airbrakes can only remove energy: they need a reliable altitude and speed early in the coast.

→

APPROACH
IMU-driven Kalman filter: accelerometer-bias state, density-scaled baro variance; linear MPC.

→

RESULT (SIMULATION)
Apogee error ≈ 12 m in 6-DOF co-simulation, with the MPC partly fed by the true state.

1 · BOOST, M < 0.9	2 · SUPERSONIC, > 4 g	3 · SUPERSONIC COAST	4 · SUBSONIC COAST
IMU + barometer	**IMU only: dead reckoning**	**IMU + GNSS velocity**	**IMU + baro + GNSS · airbrakes active**

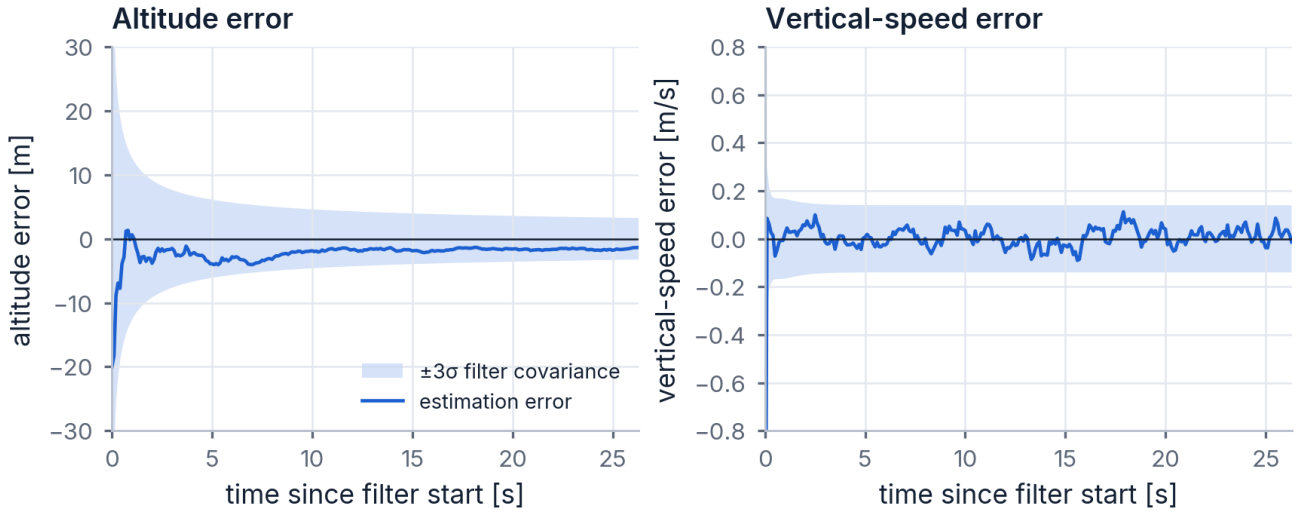
launch — sensors selected by flight regime — apogee

MY PART

- Co-developing the coasting-phase Kalman filter and the MPC chain.
- Per-step re-linearisation of the drag dynamics, now used by the MPC.
- Helping port the filter to C (FreeRTOS, ESP32-S3).



Sunspear CAD (structure team). Target apogee 9 km.



Coasting-phase filter (Python re-run). From 20 m and 2 m/s off, the error stays inside its own ±3σ bounds.

Tools MATLAB · fmincon (SQP) · C / FreeRTOS on ESP32-S3

UAV HARDWARE & FIELD TEST · ISAE-SUPAERO LOW-COST DRONE CHALLENGE · DEC 2025 – JUN 2026

Low-cost interceptor UAV: from sketch to field test

Built with a team of six for the French Army 11th Parachute Brigade: **1st place of four**.

CHALLENGE
Two identical drones able to intercept enemy drones, ≤ €3k per team.

→

VEHICLE
7-inch FPV quad, camera + ToF LiDAR in the nose, no GNSS.

→

RESULT
Field-tested with Brigade operators on 26 May 2026; team 1st place.

MY PART

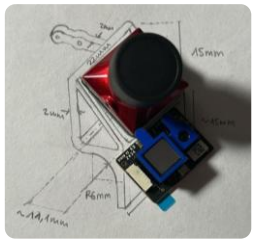
- Camera + LiDAR nose cradle: designed, drawn, 3D-printed and flown.
- Avionics integration spec: FC soft-mount, LiDAR UART, EMI.
- Betaflight config, incl. LiDAR range on the OSD.



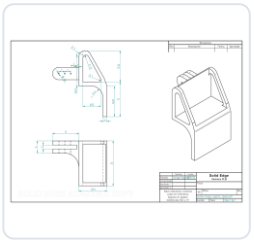
Field test (face and plate blurred).



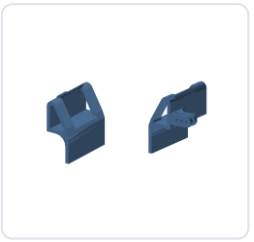
Prototype. Camera and LiDAR in the printed nose cradle.



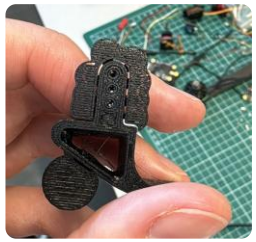
1 Sketch



2 Drawing



3 CAD



4 Printed



5 Flown

Tools Solid Edge · 3D printing · Betaflight · FPV electronics

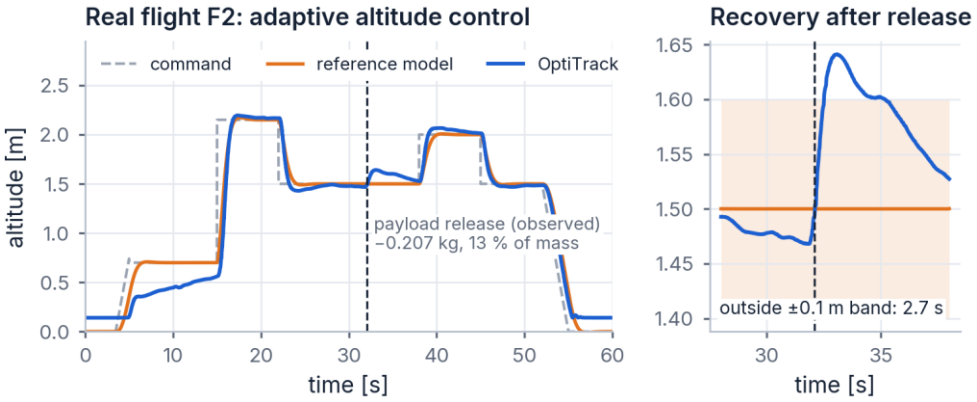
FLIGHT-TEST DATA · ISAE-SUPAERO MSC LABS · APRIL – JUNE 2026

Two flight-control labs on real quadrotors

Each controller went **simplified model → Simulink simulator → real flight**, scored against motion capture.

A · ADAPTIVE CONTROL

The drone drops 13 % of its mass mid-flight; MRAC adapts its altitude gains.

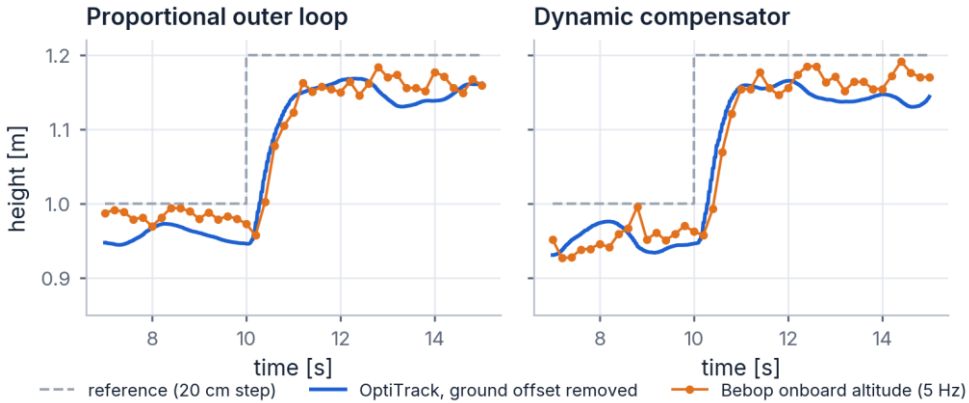


2.6–2.9 s
outside ±0.1 m of the reference model after the release, 5 flights

≈ 0.06 m
steady altitude RMSE on the plateaus

B · SENSOR LATENCY

Parrot Bebop via ROS: onboard altitude against motion capture.



≈ 0.2 s
lag of the onboard altitude (one 5 Hz sample)

5 cm
datum offset between the two altitude sources, corrected in the plot

AVAILABLE FROM MARCH 2027

Full-time role or final-year placement

ISAE-SUPAERO MSc, Aerospace Systems & Control
English C1 · French B2 · Spanish native
EASA A1/A3 + A2 remote-pilot certificates

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Tools MATLAB/Simulink · Simulink Desktop Real-Time · ROS · OptiTrack